WASHINGTON MBELEKI

SOFTWARE ENGINEER | DATA ENGINEER

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PROFILE

Software engineer with practical experience building backend services and data-backed applications, targeting the Junior C# Software Developer role at DotActiv. Key technical alignment includes Python, SQL, Java, JavaScript, TypeScript, Bash/Shell. Project work includes relational persistence with Cloudflare D1 and SQLite, caching with Cloudflare KV, third-party API integrations, workflow automation, and application features informed by operational data. Brings Python and SQL foundations alongside testing, code review, and collaborative delivery.

DATA ENGINEERING SKILLS

Languages & query	| Python, SQL, Java, JavaScript, TypeScript, Bash/Shell
Data & storage	| SQLite, Cloudflare D1, Cloudflare KV, Relational data storage, Persistence, Caching, ETL concepts
Backend & integration	| REST APIs, Flask, Node.js, Hono, Cloudflare Workers, Microservices, JWT, Third-party API integration
Workflow & engineering	| Workflow automation, TDD, JUnit 5, Object-oriented design, Git/GitHub, GitLab CI/CD, Linux, Docker, Maven, Agile

EXPERIENCE

Peer Tutor & Bootcamp Assistant | *WeThinkCode_* Oct 2025 - Present

- Facilitate hands-on sessions covering Python fundamentals, Java object-oriented design, and Git workflows for incoming cohorts.
- Conduct code reviews, help troubleshoot systems, and reinforce test-driven development and maintainable coding practices.

SELECTED PROJECTS

Automated Job Pipeline & Tailored PDF Resume Engine | *Personal Project*

- Engineered an automated lead pipeline scraping public job feeds using Python, BeautifulSoup, and backoff retry handling.
- Implemented TF-IDF cosine similarity scoring to match candidate profiles against incoming job descriptions.
- Built a Jinja2 & xhtml2pdf rendering engine to automatically generate job-tailored PDF CVs and managed workflow tracking via a Streamlit dashboard and SQLite.

MoCredit - AI Credit Scoring & Micro-Lending Mini App | *2nd Place, MTN MoMo FinTech Hackathon, 2026*

- Worked on a fintech application using a Node.js/Hono backend on Cloudflare Workers, Cloudflare D1 for persistent storage, and Cloudflare KV for score caching.
- Integrated MTN MoMo Collections (RequestToPay) and Disbursements APIs, with an n8n-orchestrated AI coaching workflow.

Thuso Health - Digital Health Passport & Smart Clinic Finder | *Top 3, G13 Health Challenge*

- Co-built an offline-first health passport and clinic finder with backend APIs using Hono/Cloudflare Workers, D1 records, R2 medical attachments, and JWT authentication.
- Developed data-informed clinic recommendations using travel-time, queue-congestion, and load-shedding/solar-power information; supported offline caching and sync-on-reconnect.

TSUMO Network - Backend Data Architecture

- Co-engineered Python-based REST API endpoints handling media-streaming metadata workflows and user request routes.
- Implemented server-side business logic to automate complex, usage-based revenue allocation and payout calculations.
- Integrated application layers with persistent relational database storage to ensure safe, validated transaction logging.

Code Clinic Booking System | *WeThinkCode_ course project*

- Built a booking application for scheduling and managing academic code clinic sessions using Python, Flask, and SQLite.

Robot World Simulation Platform

- Built a Java multiplayer client-server simulation using Maven, object-oriented design, test-driven development, Swing, and socket/network programming; refactored code in response to complexity warnings.

EDUCATION

Occupational Diploma in Software Engineering | WeThinkCode_ Oct 2025 - Expected Feb 2027

Johannesburg, South Africa | *Relevant coursework:* Software Engineering Principles, Data Structures & Algorithms, Object-Oriented Design, Concurrent Systems, CI/CD, and Database Management.

TRAINING & CERTIFICATIONS

- Data Engineering Basics for Everyone — edX, Aug 2026.
- Relational Database Management Systems — IBM, Sep 2026.
- Generative AI for Software Engineering — WeThinkCode_; prompt engineering, AI-assisted refactoring, test-suite generation, and LLM API integrations.

ROLE-FIT SUMMARY

> *Role-fit summary for Junior C# Software Developer at DotActiv: Current project evidence is strongest in database-backed applications, API integration, caching, and data-informed backend logic matching Python, SQL, Java, JavaScript, TypeScript, Bash/Shell. Dedicated end-to-end ETL pipeline operations represent a continuous development priority.*

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Strategic Brief: Candidate Positioning & Value Proposition

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POSITIONING

Software engineer targeting entry-level data engineering roles, with hands-on project evidence across relational storage, API integration, caching, and data-informed application features. MoCredit and Thuso Health show how I connect data and services to support application workflows. My next step is to extend this foundation into clearly demonstrated ingestion, transformation, data-quality checks, and repeatable pipeline operation.

TARGET ROLES & WORKING SETUP

- **Roles:** Entry-Level / Junior Data Engineer; Data Engineering Intern or Trainee; Graduate Data Engineer; Junior ETL Developer; backend roles with substantial SQL and data-processing work.
- **Environment:** A team with mentorship, code review, clear data-quality expectations, and opportunities to build and operate small pipelines.
- **Application Strategy:** Prioritize roles that value Python, SQL, backend fundamentals, and learning potential. Treat Spark, Airflow, dbt, and cloud data warehouses as learning targets unless separate hands-on evidence is available.

VALUE PROPOSITION

- **Data-Backed Application Workflows:** MoCredit combines D1 persistence, KV caching, financial API integrations, and workflow automation.
- **Operational Data Features:** Thuso Health recommendations use travel-time, queue-congestion, and energy-availability information, alongside offline caching and sync considerations.
- **Engineering & Backend Discipline:** TSUMO Network, Code Clinic Booking, and peer tutoring provide evidence of API/service work, Python/SQLite, debugging, testing, code review, and teamwork.

PROOF POINTS

Project	What It Demonstrates	Stack
Job Pipeline Engine	Scraping, TF-IDF semantic matching, SQLite tracking, and dynamic PDF generation.	`Python, BeautifulSoup, TF-IDF, SQLite, Streamlit, xhtml2pdf`
MoCredit	Persistent storage, caching, external API integration, workflow automation; 2nd place at MTN MoMo FinTech Hackathon 2026.	`Node.js, Hono, Workers, D1, KV, n8n`
Thuso Health	Data-informed recommendations, offline-first behavior, records and attachment storage.	`Hono, Workers, D1, R2, JWT`
TSUMO Network	Backend services, REST endpoints, revenue calculations, persistence integration.	`Python, APIs, persistent storage`
Code Clinic	Database-backed booking workflow.	`Python, Flask, SQLite`

STRATEGIC BRIEF: GAPS & DEVELOPMENT PLAN

KNOWN GAPS & HOW I AM CLOSING THEM

- **No standalone end-to-end ETL project yet:** Build a public-data pipeline that ingests a CSV or API source, validates and transforms records with Python, and loads them into SQLite or PostgreSQL.
- **SQL depth needs visible proof:** Add examples for joins, aggregations, CTEs, window functions, null handling, deduplication, data-quality checks, and query optimization.
- **Reliability and orchestration:** Add repeatable runs, logging, environment-based configuration, error handling, and safe reruns. Add scheduling after core pipeline is stable.

FIRST 90 DAYS (PROPOSED)

Period	Focus	Evidence to Produce
Days 1-30	Build a complete Python + SQL pipeline	Public CSV/API ingestion, validation/transformation, relational database loading, unit tests and sample outputs.
Days 31-60	Improve SQL and pipeline reliability	Joins, aggregations, window functions, data-quality checks, logging, error handling, and repeatable runs.
Days 61-90	Polish and present the project; apply consistently	Clear repository, data-flow diagram, project walkthrough, interview practice, and tailored applications.